John Duncan

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https://johndunking.github.io/jduncan-portfolio/

SUMMARY

Special education practitioner and software developer with 10 years in K–5 education, combining IEP and compliance expertise, instructional coaching, and professional development design with the ability to build the technology that supports them. Independently designed and deployed a full-stack educational workflow platform for instructional planning and scheduling (Python/FastAPI, PostgreSQL, Docker on Google Cloud Run), and built two native iOS and Unity learning applications — a speech-driven K–3 literacy program and an iPad application for individualized visual supports architected to run fully offline with no student accounts and no data transmission. Brings practitioner insight to product and accessibility decisions, collaborates directly with site leadership and educators, and owns technical implementation end to end.

PROFESSIONAL EXPERIENCE

Education Specialist

Alexander Twilight College Preparatory Academy (Aspire Public Schools) — Sacramento, CA · Jul 2023 – Present

- Coordinate IEP development, service planning, progress monitoring, and multidisciplinary collaboration to support timely, legally compliant special education services.
- Coach instructional aides and collaborate with general education teachers to strengthen implementation of individualized academic and behavioral supports.
- Design and facilitate professional learning for educators focused on instructional practices, behavioral supports, and equitable student access, as a member of the CV Equity Leadership Team.
- Mentor induction and early-career teachers through instructional feedback, collaborative planning, and professional support (Induction and SCOE New Teacher Mentor).
- Collaborate with site leadership to strengthen inclusive practices and improve tiered intervention systems.

After School Director

Aspire Capitol Heights Academy (Aspire Public Schools) — Sacramento, CA · Feb 2019 – Feb 2022

- Directed daily after-school program operations and supervised, coached, and evaluated a team of instructional staff, including performance feedback and professional development.
- Implemented and managed the Bricks 4 Kidz enrichment program to expand STEM-based student engagement.

Game Programmer (Internship)

[P1] Virtual Civilization Initiative — Remote, CA · Nov 2023 – Nov 2024

- Debugged and supported the team with code adjustments in C# using the Unity project manager.

Software Developer / AI Developer (Contract)

DataAnnotation — Remote, CA · Dec 2023 – Mar 2024

- Created a Python algorithm to sort and process text files to specification.
- Tested and debugged AI model responses for quality and correctness.

SELECTED PROJECTS

- **SPEDstack — Educational Workflow Platform** — Independently developed an educational workflow platform supporting the daily work of educators — importing weekly instructional plans, resolving curriculum and pacing, building co-teaching and support-staff schedules, coordinating administrative follow-through, and generating the planning documents teams use each week. Designed so staff work from the spreadsheets they already use rather than learning a new system. Python, FastAPI, PostgreSQL, Docker, and Google Cloud Run.

- **ReadingAvatar — K–3 Literacy Learning Application** — Designed and built a K–3 literacy application (Swift/iOS, migrated to Unity/C#) in which a student's reading aloud drives the gameplay across phonics, fluency, vocabulary, and comprehension. Implemented on-device speech recognition and a mastery estimator that weights evidence by how it was gathered, with design decisions recorded as architecture decision records; ~1,100 automated tests.
- **VisualAvatar — Individualized Visual Supports (iPad)** — Built a native iPad application delivering individualized visual supports — visual cues, first-then boards, choice boards, and task sequences — for students with disabilities. Architected it to run fully offline and on-device with no student accounts and no data transmission, enforced by an automated architecture check that fails the build if networking, analytics, or third-party dependencies enter shipping code. Swift and Unity/C# with module boundaries enforced at compile time; ~1,065 automated tests plus documented on-device verification.
- **Classroom Web Tools (open source)** — Published a set of small React and Phaser classroom tools — math worksheet generation, handwriting tracing, student intervention matching, and classroom management — several deployed live.

TECHNICAL & PROFESSIONAL SKILLS

Software Engineering: Python (FastAPI, Pydantic, SQLAlchemy, pytest); Swift / SwiftUI (iOS, SPM, XCTest); C# / Unity (asmdef modules, EditMode/PlayMode tests); JavaScript / React (CRA, React Router, Bootstrap, GSAP); Cloud deployment: Google Cloud Run, Cloud SQL (Postgres), Docker; LLM/API integration (Anthropic, OpenAI); Google Workspace / Sheets API automation

Product Engineering: Accessibility & assistive-technology design (individualized visual supports, speech-driven interaction); Privacy-by-design for student data (fully on-device architecture, no accounts, no data transmission); Product scoping & requirements definition (documented scope and design decisions); Software architecture & architecture decision records (ADRs); Automated testing & release verification (pytest, XCTest, Unity Test Framework); Stakeholder collaboration with site leadership, educators, and families

Education Leadership: IEP development & special education compliance (IDEA/CA); Instructional coaching (aides & gen-ed teachers); Professional development design & facilitation; Data disaggregation & equity-gap analysis (Dashboard/i-Ready/mCLASS); Program management & staff supervision

EDUCATION

M.A. in Education — Alder Graduate School of Education (Jun 2023)

B.S. in Kinesiology — Sacramento State University (2020)

CREDENTIALS

Education Specialist Instruction Credential (expires Jun 2030)

Single Subject Teaching Credential (expires Jun 2030)

Preliminary Administrative Services Credential — in progress, expected Jun 2027